

U.S. Cities Weather Analysis

Data Analysis | January - December 2025

Cities: New York . Los Angeles . Chicago . Houston . Phoenix

About This Report

This report analyses daily weather data for five major U.S. cities across all of 2025. Using temperature and precipitation records, we explore seasonal trends, city comparisons, extreme weather events, and stability patterns. All findings are presented visually so anyone - technical or not - can follow along.

What's Inside

1. Temperature Overview	Min/max spread, city-level boxplots
2. Monthly Distributions	Per-city boxplots broken down by month
3. Yearly Trends	Temperature and precipitation timelines
4. City Comparisons	Heatmap, volatility, hottest & coldest rankings
5. Key Findings	Extreme events, seasonal patterns, stability analysis

Key Insights

- Phoenix peaks at ~38degC in summer - the hottest city by far.
- Chicago has the sharpest seasonal swing, dropping below 0degC in winter.
- Los Angeles is the most stable city year-round (lowest temperature std).
- July-August are the harshest months when combining heat and precipitation.
- Houston records the most high-precipitation days (>20mm) among all cities.

1. Temperature Overview

The scatter plot below shows how minimum and maximum daily temperatures move together. Each dot is one day - the tight diagonal line tells us that on hot days, even the lows stay warm, and on cold days, everything drops together.

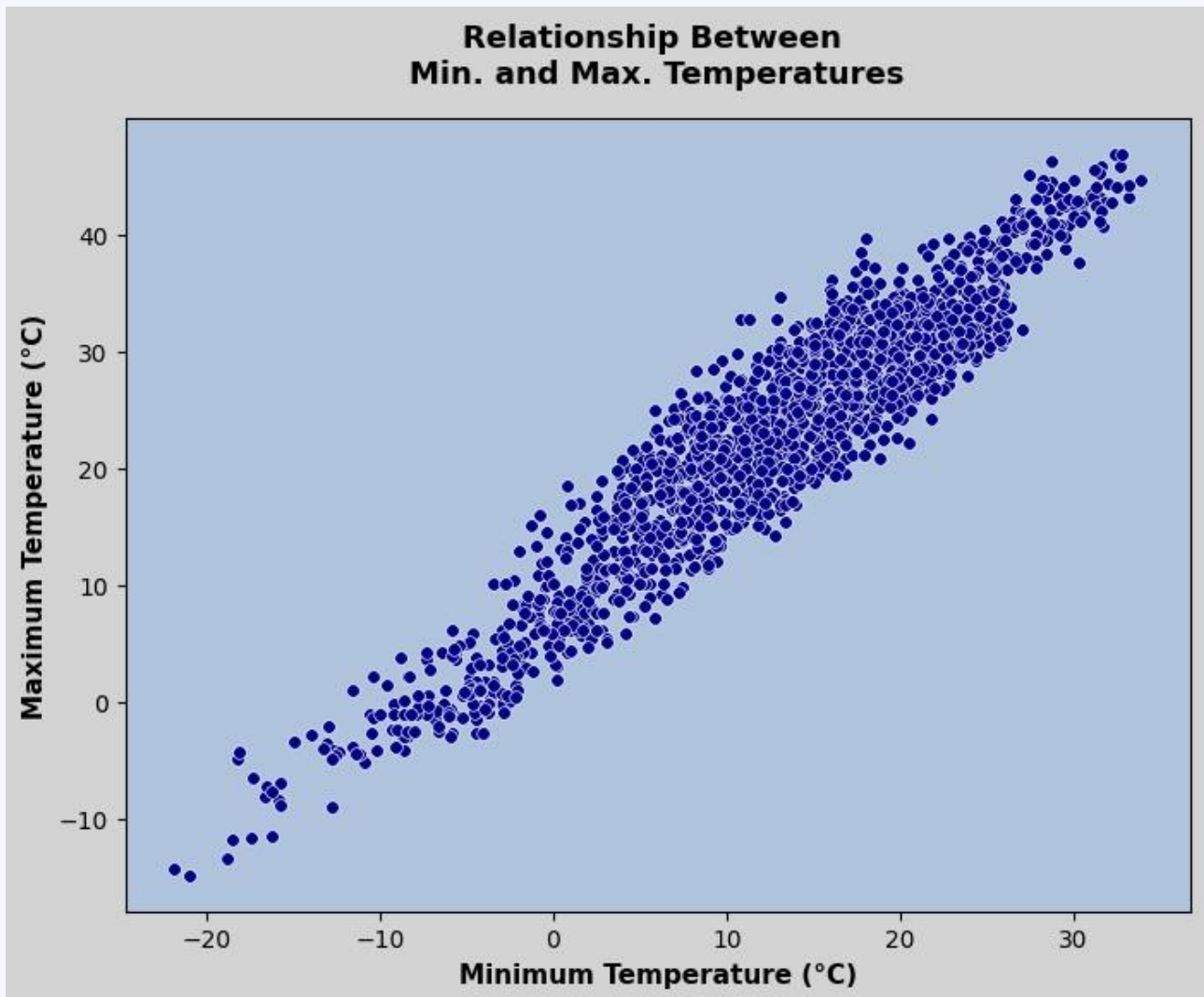
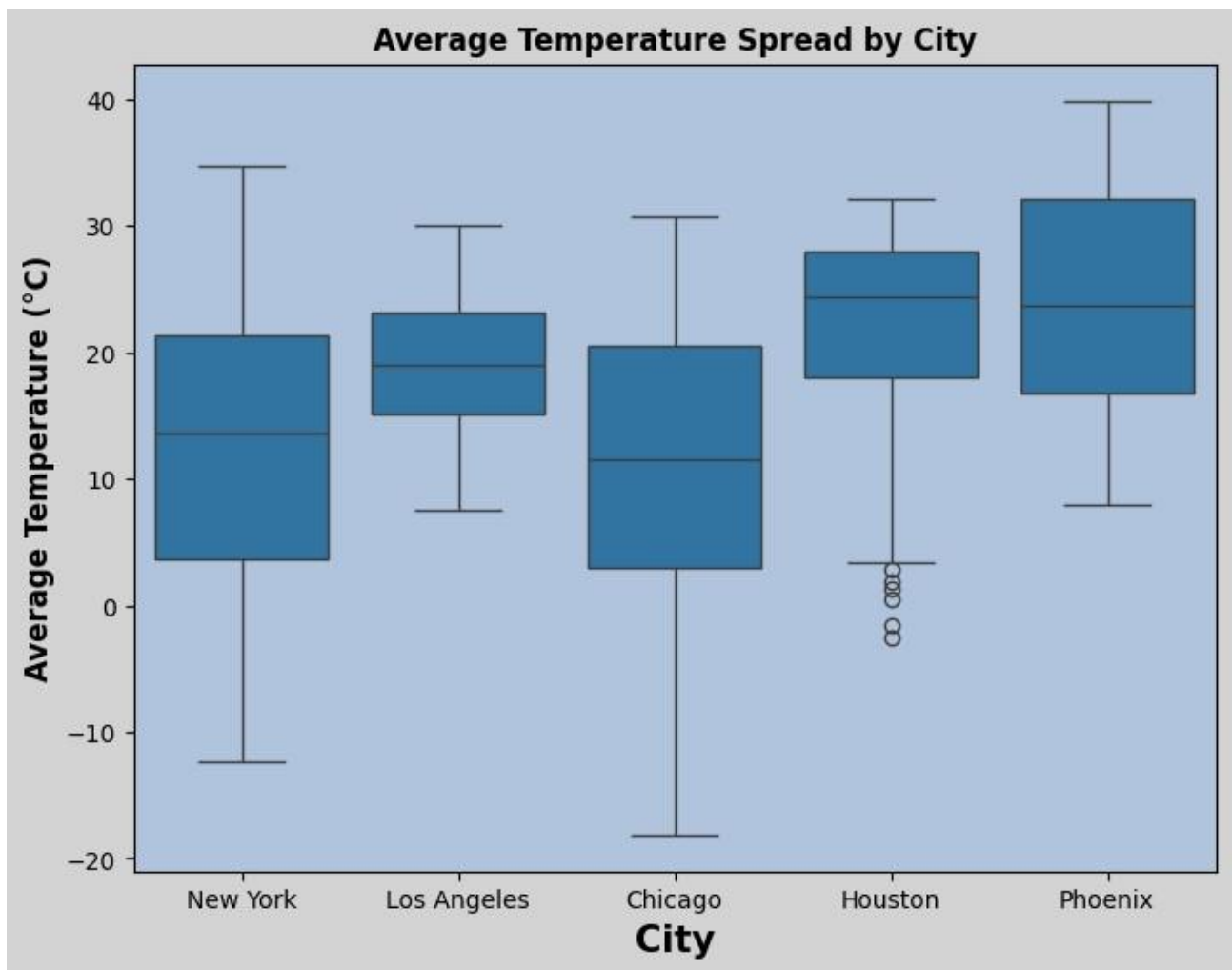


Fig 1. Relationship between daily minimum and maximum temperatures across all cities.

The boxplot compares the full-year temperature spread for each city. Phoenix sits highest (median ~28degC), while Chicago sits lowest and shows the widest range - meaning its weather swings the most unpredictably.



2 Monthly Temperature Distributions

Each panel below shows one city's month-by-month temperature spread as a boxplot. Wider boxes mean more day-to-day variation in that month. Phoenix shows extreme summer peaks; Chicago shows dramatic winter lows.

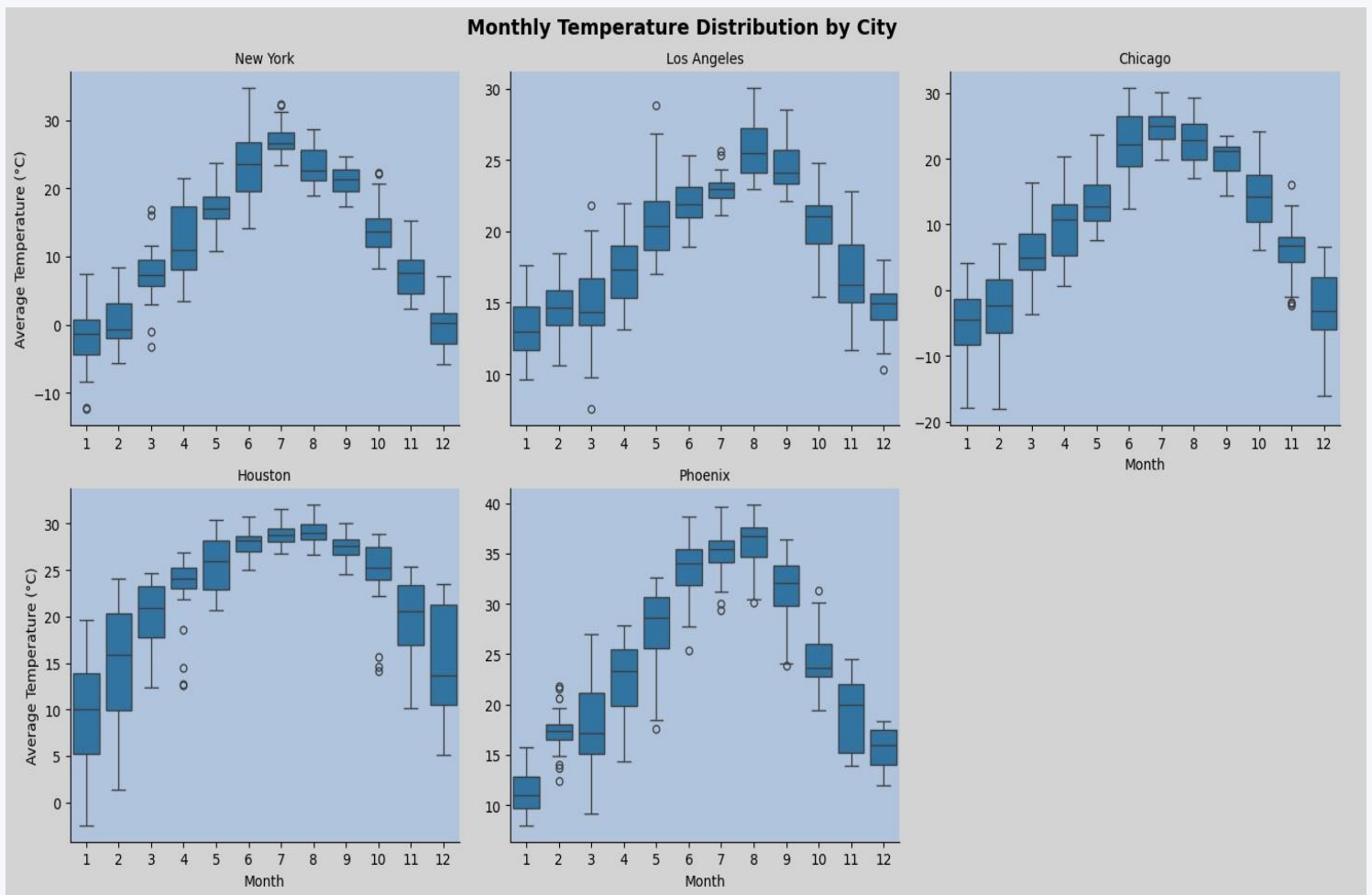


Fig 3. Monthly temperature boxplots per city - each column is a month (1=Jan, 12=Dec).

3. Yearly Trends

The line chart tracks each city's average monthly temperature through the year. All cities follow a clear summer peak, but Phoenix (orange) towers above the rest in July-August, while Chicago (green) dips furthest in January-February.

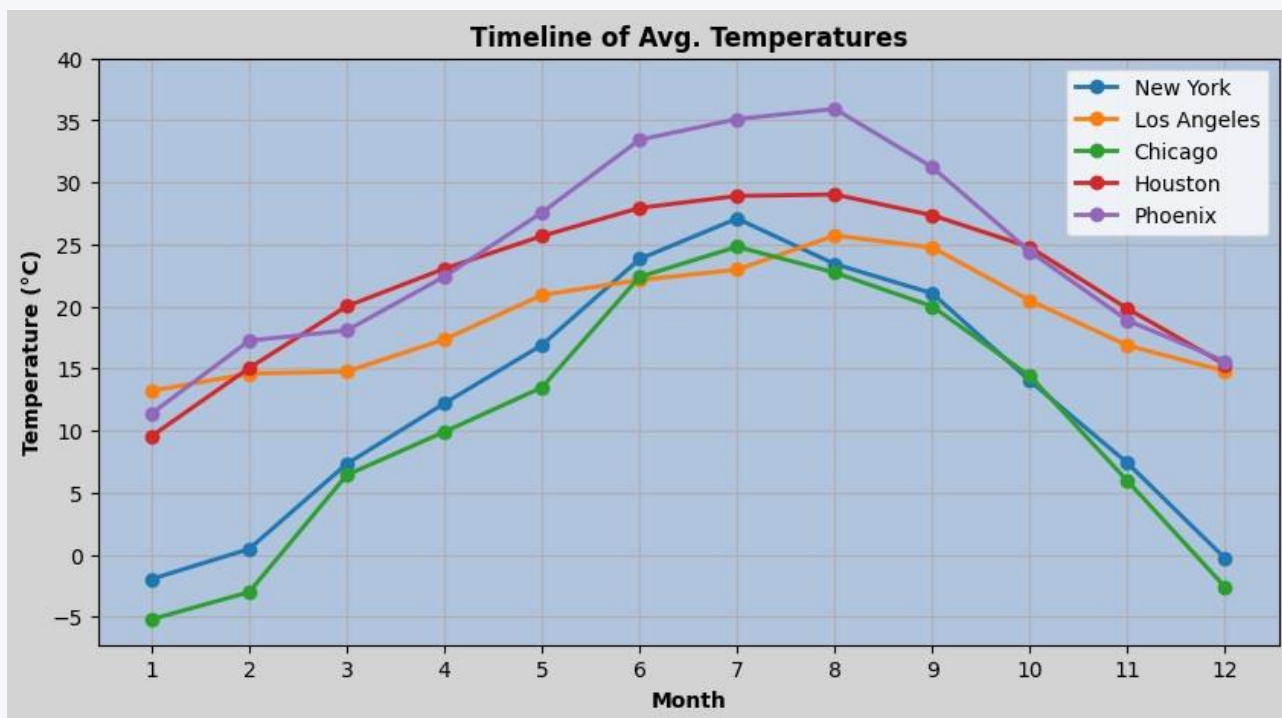


Fig 4. Monthly average temperature timeline for all five cities.

Precipitation follows a less uniform pattern. New York and Chicago see more rainfall in mid-year, while Los Angeles stays remarkably dry throughout summer - consistent with its Mediterranean climate.

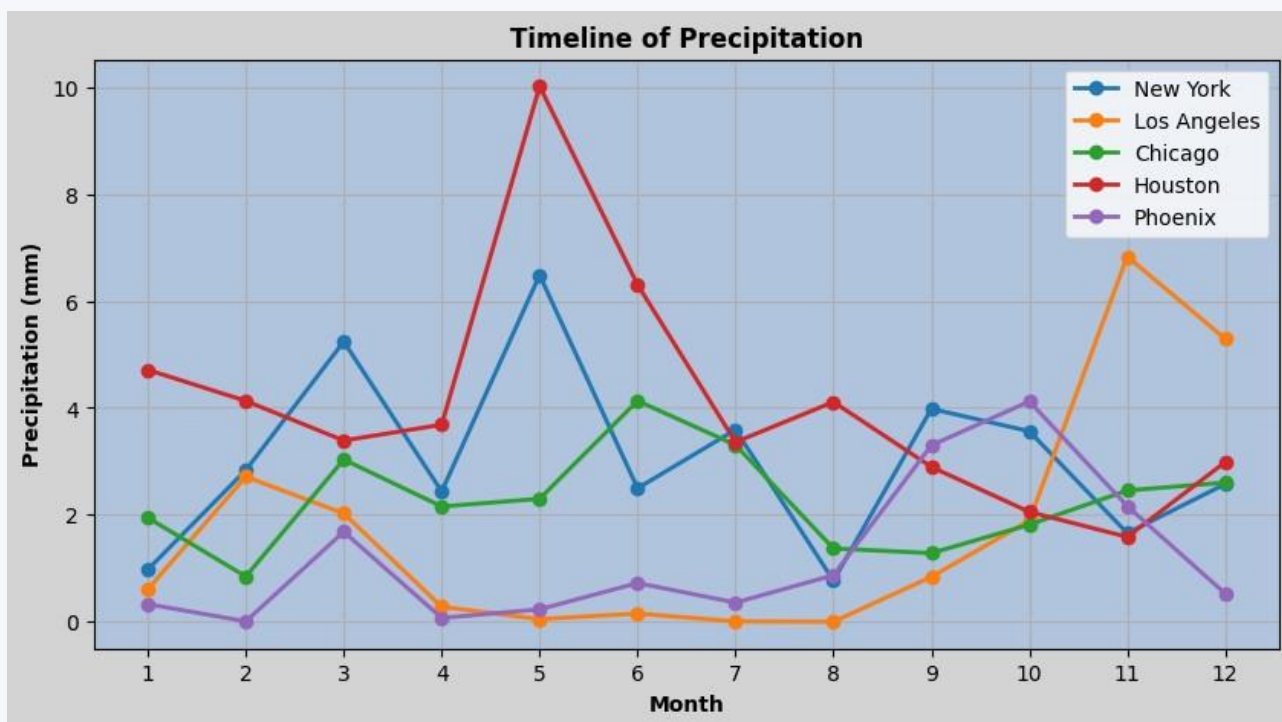


Fig 5. Monthly average precipitation (mm) timeline for all five cities.

4. City Comparisons

The heatmap gives a single-glance view of all cities across all months. Deep red means hot, deep blue means cold. Phoenix's entire summer block burns red; Chicago's winter block goes deep blue.

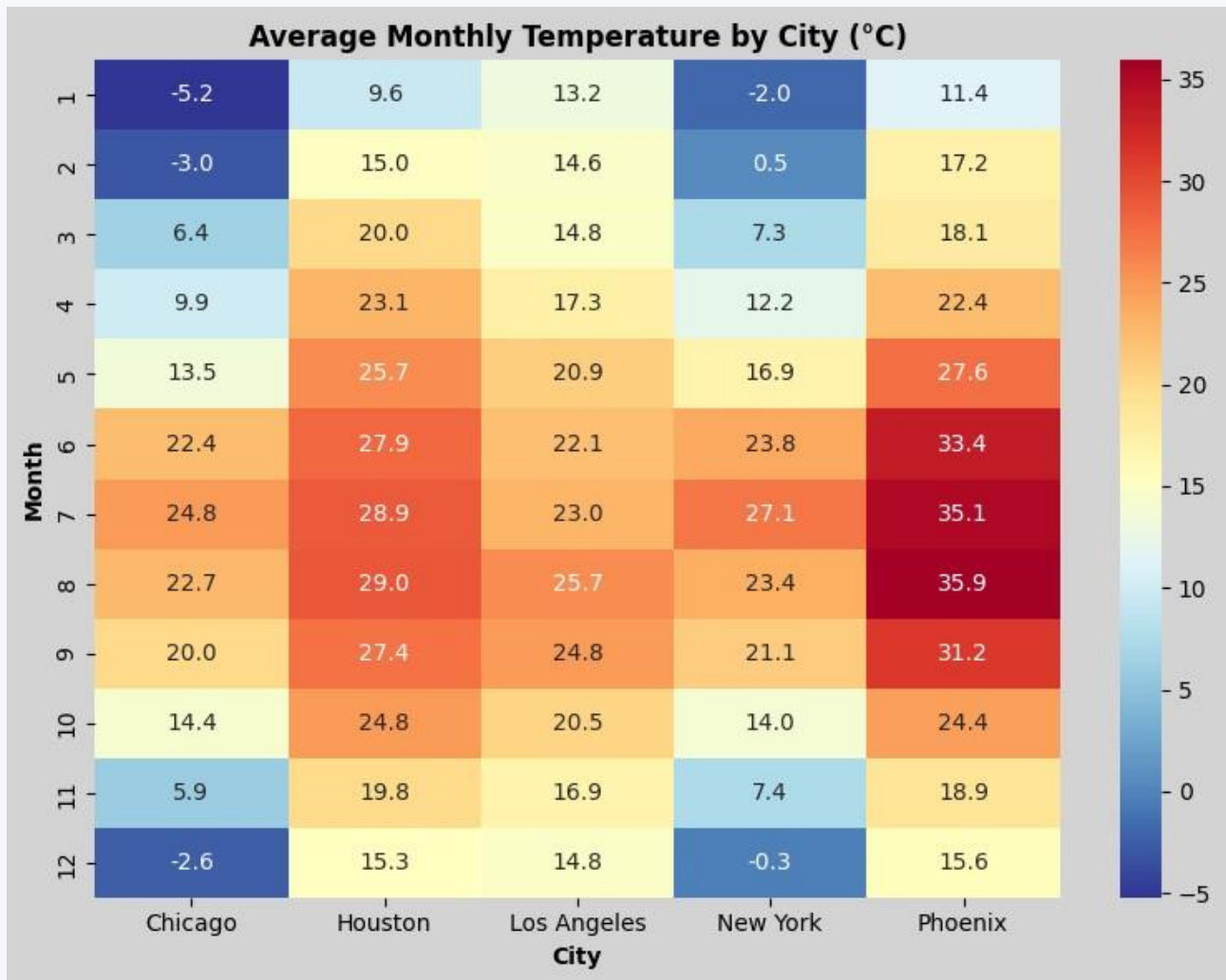


Fig 6. Heatmap of average monthly temperatures (degC) - red = hot, blue = cold.

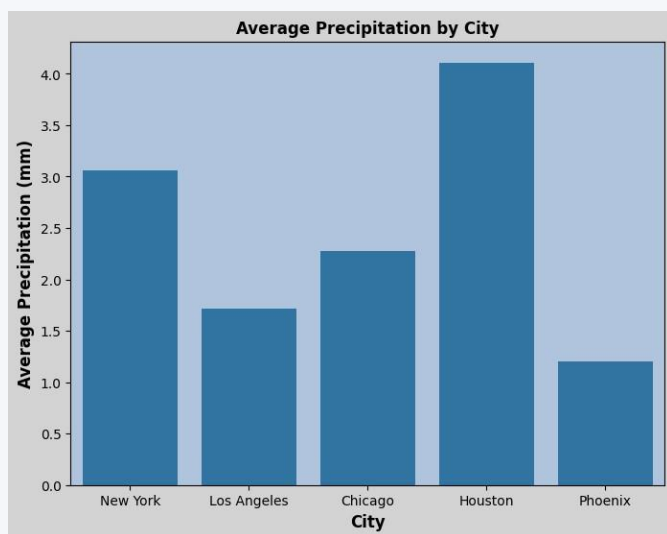


Fig 7. Average daily precipitation by city.

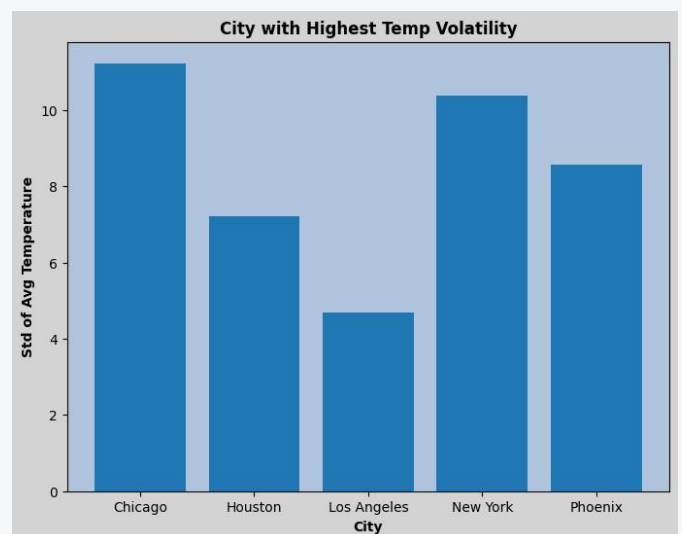


Fig 8. Temperature volatility (std) by city.

Key Insights

- **Houston** gets the most rain on average - nearly double that of Los Angeles.
- **Chicago** has the highest temperature volatility (std ~10.5degC) - most unpredictable.
- **Los Angeles** has the lowest volatility - weather barely changes month to month.
- **Phoenix** is hottest overall but gets very little precipitation.

5. Key Findings

Most Extreme Seasons

Chicago

Chicago has the largest gap between its hottest and coldest seasonal averages - nearly 30degC difference between summer and winter. This makes it the hardest city to dress for year-round.

Harshest Month

July / August

When combining heat and precipitation into a single 'harshness score', July and August rank highest. These months bring both peak temperatures and elevated rainfall across most cities.

Most Stable City

Los Angeles

Los Angeles has the lowest standard deviation of average temperature (~7degC), meaning its weather barely fluctuates. You could wear the same outfit most of the year.

Phoenix - The Outlier

Extreme heat, mild winters

While most cities are coldest in winter and hottest in summer, Phoenix stands out because its winters are actually pleasant (~12degC), while its summers are brutal (~38degC). The seasonal swing is extreme but skewed - it's not cold, just less hot.

Longest Heatwave

Phoenix leads

Phoenix records the longest consecutive streak of days above 35degC. Houston follows, but its high humidity makes even moderate temperatures feel more intense.

Data source: Open-Meteo API | Period: Jan 1 - Dec 31, 2025 | Cities: New York, Los Angeles, Chicago, Houston, Phoenix |

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